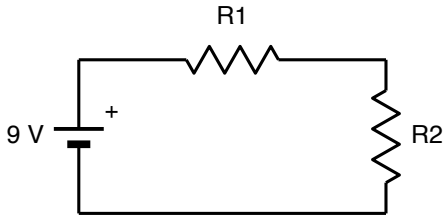
Station 1: Multimeter: voltage, current, resistance

15 min · 20 points

Name: _____ Start time: _____

At this station: 9 V battery with snap leads, breadboard, jumpers, multimeter, resistors R1 and R2, two resistors marked “1M” (R3, R4).

Tasks



1. Set the meter to Ω. Measure R1 and R2 **out of the circuit**. Do not touch both probe tips.
2. Build the series circuit in the figure. Measure Vs (battery, circuit connected), V1 (across R1) and V2 (across R2). Write the meter range next to each reading.
3. Check the voltage law: V1 + V2 = Vs?
4. Measure the current I. Break the circuit and put the meter **in series** (red lead in the mA jack). Then move the red lead back to VΩ.
5. Calculate I = V1 / R1 with your **measured** R1. Compare it with your ammeter reading (percent difference).
6. Measure R3 and R4 (1 MΩ each) with the ohmmeter. Swap R1 and R2 for R3 and R4. Predict V4 for an ideal meter from your measured R3, R4. Measure V4.
7. Your V4 is low. Explain why. Use your reading and your measured R3, R4 to estimate the meter’s input resistance.

Hand in: All readings with units and ranges, the KVL check, the percent difference, the meter-loading explanation. **App:** N network solver: type the parts, read the node voltages and the current.

R1, R2 (ohmmeter)		Ω
Vs, V1, V2		V
V1 + V2 vs Vs		
I (ammeter)		mA
I = V1/R1; % difference		mA, %
V4 predicted / measured		V
Why V4 is low; meter resistance		MΩ

Units and significant figures on every number. Write the meter range next to each reading.

Supervisor sign-off: _____ Score: ____ / 20

Station 2: Mystery resistor: color code and voltage only

15 min · 20 points

Name: _____ Start time: _____

At this station: Four loose resistors A, B, C, D; known resistor Rk (value on its tag); mystery resistor Rx (bands covered with tape); 9 V battery; breadboard; jumpers; multimeter (volts only for Part B).

Tasks

1. **Part A.** Read the color bands of A, B, C and D. Write each value with its tolerance.
2. **Part B (voltage only).** Do **not** use the Ω or mA setting. Do not lift the tape.
3. Build Rk and Rx in series across the battery. Draw your circuit.
4. Measure Vk (across Rk) and Vx (across Rx), and Vs with the circuit connected.
5. Calculate Rx = Rk × Vx / Vk. Give 3 significant figures.
6. Give the range of Rx that the ±5% tolerance of Rk allows.
7. **Part C.** Ask for permission. Then measure Rx with the ohmmeter. Give the percent error of your Part B value.

Hand in: Part A values, a schematic, Vk, Vx, Vs, Rx with sig figs, the range, the percent error. **App:** m mystery resistor (type Rk, Vk, Vx); c color code (type the bands, one letter each).

A, B, C, D values		Ω
Schematic (draw)		
Vs, Vk, Vx		V
Rx = Rk × Vx / Vk		kΩ
Range from Rk tolerance		kΩ
Rx by ohmmeter; % error		kΩ, %

Units and significant figures on every number. Write the meter range next to each reading.

Supervisor sign-off: _____ Score: ____ / 20

Station 3: Build a target resistance

15 min · 20 points

Name: _____ Start time: _____

At this station: Resistors: two 1.0 kΩ, one 2.2 kΩ, one 3.3 kΩ, one 4.7 kΩ (read the bands yourself). Breadboard, jumpers, multimeter.

Tasks

1. Read every resistor's value. Measure each one with the ohmmeter.
2. Build **5.50 kΩ**. Use at most 2 resistors. Measure it.
3. Build **1.32 kΩ**. Use exactly 2 resistors. Measure it.
4. Build **1.82 kΩ**. Use exactly 3 resistors. Measure it.
5. Build the value **closest to 2.50 kΩ**. Any number of resistors. Give the percent error of your nominal value.
6. What are the largest and smallest values you can make with all 5? (Calculate only.)
7. For each build: draw it, give the nominal value, the measured value and the percent difference. Raise your hand for a sign-off on the 1.82 kΩ build.

Hand in: A drawing, nominal R, measured R and % difference for each build; the largest and smallest values. **App:** h target finder: type the parts, then = and the target.

5.50 kΩ: drawing, nominal, measured		kΩ
1.32 kΩ		kΩ
1.82 kΩ (sign-off)		kΩ
closest to 2.50 kΩ; % error		kΩ, %
all 5: largest / smallest		kΩ, Ω

Units and significant figures on every number. Write the meter range next to each reading.

Supervisor sign-off: _____ Score: ____ / 20

Station 4: Make two LEDs equally bright

15 min · 20 points

Name: _____ Start time: _____

At this station: 9 V battery, breadboard, jumpers, multimeter, two red LEDs, one blue LED, resistors 330, 470, 560, 680 Ω and 1.0 kΩ.

Tasks

1. Assume Vf(red) = 2.0 V and Vf(blue) = 3.0 V. Aim for about 10 mA. Stay at or below 15 mA.
2. **Part A.** Light the two red LEDs **in series** with one resistor. Show your resistor calculation. Measure V across the resistor. Calculate the current.
3. **Part B.** Put the red LED and the blue LED **in parallel** branches. Give each branch its own resistor so both currents are equal (within 15%). Show both calculations.
4. Measure V across each branch resistor. Calculate both currents. Raise your hand for a sign-off.
5. **Part C.** Now use one shared 470 Ω resistor for both parallel LEDs. What happens? Explain.
6. Equal current in two colors may not look equally bright. Explain why in one sentence.

Hand in: Resistor choices with calculations, measured voltages and currents, the Part C explanation. **App:** I LED resistor: type the supply, Vf, the LED count and the current.

Part A: R chosen; V_R; I		Ω, V, mA
Part B red: R; V_R; I		Ω, V, mA
Part B blue: R; V_R; I		Ω, V, mA
Part C: what happens and why		
Why colors look different		

Units and significant figures on every number. Write the meter range next to each reading.

Supervisor sign-off: _____ Score: ____ / 20

Station 5: Power supplied by a source (and a USB label)

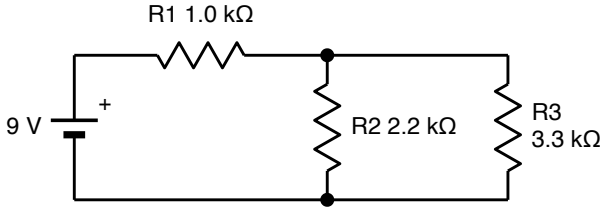
15 min · 20 points

Name: _____ Start time: _____

At this station: 9 V battery, breadboard, jumpers, multimeter (**volts only**), R1 = 1.0 kΩ, R2 = 2.2 kΩ, R3 = 3.3 kΩ. A USB charger label (below).

Tasks

1. Build the circuit in the figure. Use only voltage readings.
2. Measure Vs (battery, connected), V1 (across R1) and V23 (across R2 and R3).
3. Calculate the current from the battery. Explain why V1/R1 is the battery current.
4. Calculate the power supplied by the battery.
5. Calculate the power used by R1, R2 and R3. Show that they add up to the power supplied.
6. **USB label.** Read the label. Give the power of each output level.
7. A phone battery is 5000 mAh at 3.7 V. Charging is 85% efficient. How long does it take at the 9 V level? At the 5 V level?



CHARGER LABEL Model PD-20 Input: 100–240 V ~ 50/60 Hz 0.6 A Output: 5.0 V ⎓ 3.0 A, 9.0 V ⎓ 2.22 A, 12.0 V ⎓ 1.67 A (20 W max)

Hand in: Three voltages, the current, the power supplied, the three part powers and their sum, the USB answers. **App:** N network solver (power of every part); z USB charge time.

Vs, V1, V23		V
I = V1/R1		mA
Power supplied		mW
P1, P2, P3; sum		mW
USB: power at 5 V, 9 V, 12 V		W
Charge time at 9 V; at 5 V		h

Units and significant figures on every number. Write the meter range next to each reading.

Supervisor sign-off: _____ Score: ____ / 20

Station 6: Troubleshoot a faulty circuit

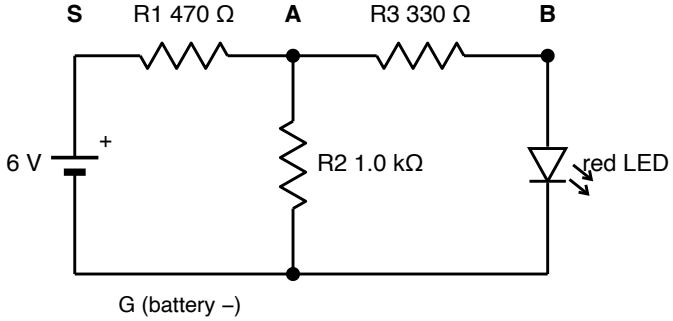
12 min · 20 points

Name: _____ Start time: _____

At this station: A built circuit (figure) on a 4 × AA pack (6 V), multimeter. Spare jumpers and the resistor values in the figure. The circuit has **one** fault, or none.

Tasks

1. Do not rebuild. Measure first.
2. Measure the pack voltage Vs. Then measure V at points A and B and across R1. Black lead on G.
3. Write the voltages you **expect** for a good circuit (LED Vf = 2.0 V). Compare.
4. Name the fault and where it is. Say which reading proves it.
5. Fix it with one change. Measure A and B again. Raise your hand for a sign-off.
6. Explain: if a series part is open, which part has the full supply voltage across it?



Hand in: Expected and measured voltages, the fault, the proof, the fix, the re-measured voltages. **App:** N network solver: type the good circuit (D3-0 2 = a 2 V LED) and compare with your readings.

Expected V_A, V_B (good circuit)		V
Measured Vs, V_A, V_B, V_R1		V
Fault and its place		
Reading that proves it		
Fix; V_A, V_B after		V
Open series part question		

Units and significant figures on every number. Write the meter range next to each reading.

Supervisor sign-off: _____ Score: ____ / 20

Station 7: Logic circuit from a truth table

20 min · 20 points

Name: _____ Start time: _____

At this station: 3 × AA pack (4.5 V), breadboard, jumpers, multimeter, 74HC00 (NAND), 74HC08 (AND), 74HC32 (OR), three SPDT slide switches, red LED, 1.0 kΩ resistor, pin chart below.

Tasks

1. **No battery yet:** find each switch's middle (common) pin by continuity. A wrong pin shorts the pack. Every chip: pin 14 to +4.5 V, pin 7 to –. Switch common to the gate input, one side to +, the other to –. Tie unused inputs to –.
2. The output drives the LED through the 1.0 kΩ resistor. LED on = 1.
3. **Part A.** Build a circuit for Table A. Write its Boolean expression first. Use the 74HC08 and the 74HC32.
4. **Part B.** Build Table B with the 74HC00 **only** (NAND gates). Draw the gates and pin numbers.
5. Test every row. Get a sign-off on each part. Measure the output voltage for a 1 and for a 0.

Table A

A	B	C	F
0	0	0	0
0	0	1	1
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Table B

A	B	F
0	0	0
0	1	1
1	0	1
1	1	0

Pins (all three chips)
Gate 1: in 1, 2 → out 3
Gate 2: in 4, 5 → out 6
Gate 3: in 9, 10 → out 8
Gate 4: in 12, 13 → out 11
Pin 14 = +, pin 7 = –. Notch to the left: pin 1 is the bottom-left pin. Pins count counter-clockwise (seen from the top).

Hand in: Both expressions, gate drawings with pin numbers, the HIGH and LOW output voltages. **App:** t truth table: type the output column (rows 000 to 111) and read F.

Table A expression		
Table A drawing with pins		
Table B expression and gate name		
Table B drawing (NAND only)		
Output V for 1 and for 0		V

Units and significant figures on every number. Write the meter range next to each reading.

Supervisor sign-off: _____ Score: ____ / 20

Station 8: Wire a 3-way switch

12 min · 20 points

Name: _____ Start time: _____

At this station: 9 V battery, breadboard, jumpers, multimeter, two SPDT slide switches, red LED, 1.0 kΩ resistor.

Tasks

1. Use the meter's continuity mode. Find the common pin of each switch. Explain your test.
2. Wire the LED so that **either** switch turns it on or off, from any state (a stairway light). Draw your schematic first.
3. Fill in the table: LED state for all four switch positions.
4. With the LED on, measure V across the resistor. Calculate the LED current.
5. Which logic gate is your table? What changes if you swap the two traveler wires?
6. How many switches, and of what kind, would you need for three places?

Hand in: Common-pin method, schematic, the 4-row table, V_R and I, the gate name, the three-place answer. **App:** t truth table (type the LED column); I LED current.

Common pin: how you found it		
Schematic (draw)		
Table: S1 S2 LED (4 rows)		
V_R; I		V, mA
Gate; travelers swapped		
Three places		

Units and significant figures on every number. Write the meter range next to each reading.

Supervisor sign-off: _____ Score: ____ / 20

Station 9: Design a voltage divider

15 min · 20 points

Name: _____ Start time: _____

At this station: 9 V battery, breadboard, jumpers, multimeter, resistors: three 1.0 kΩ, one each of 2.2, 3.3, 4.7 kΩ. A 10 kΩ resistor marked “LOAD”.

Tasks

1. Target: **Vout = Vs/3 ± 5%** (3.0 V from a 9.0 V battery). Measure Vs: a fresh battery is above 9 V.
2. **Part A.** Design a divider for 3.0 V with no load. Show the calculation. Build it. Measure Vs and Vout.
3. Give the expected Vout for your measured Vs. Give the percent error.
4. **Part B.** Connect the LOAD across Vout. Measure Vout again. Explain the change.
5. **Part C.** Design a new divider that gives 3.0 V ± 5% **with the LOAD connected**. Calculate first. Build it. Raise your hand for a sign-off.
6. Calculate the current your Part C divider takes from the battery.

Hand in: Both designs with calculations and drawings, all measured voltages, the loading explanation, the current. **App:** d voltage divider (type the load in the R2 box as 3.3k/10k = parallel).

Part A: R_top, R_bottom; calculation		kΩ
Part A: Vs, Vout; % error		V, %
Part B: Vout with LOAD; why		V
Part C: design and calculation		kΩ
Part C: Vout with LOAD (sign-off)		V
Battery current		mA

Units and significant figures on every number. Write the meter range next to each reading.

Supervisor sign-off: _____ Score: ____ / 20

Station 10: Diode and LED: orientation and forward voltage **12 min · 20 points**

Name: _____ Start time: _____

At this station: 9 V battery, breadboard, jumpers, multimeter, LEDs P, Q, R (legs cut to the same length), diode S (small glass body), resistors 1.0 kΩ and 10 kΩ.

Tasks

1. Set the meter to diode test. For P, Q, R and S: find the anode. Record the reading both ways.
2. From the readings, name the likely color (or type) of each part. Which one is silicon?
3. Build: battery → 1.0 kΩ → LED P → battery. Measure V_LED and V_R. Calculate I.
4. Swap the 1.0 kΩ for 10 kΩ. Measure V_LED and I again. Does Vf depend on the current?
5. Put diode S in place of the LED, first forward, then backwards. Measure V_S and V_R both ways. Explain the readings.

Hand in: Diode-test table (both ways), anode legs, part names, V_LED, V_R and I for both resistors, the diode readings. **App:** I LED tool works for a silicon diode too (Vf = 0.7).

Diode test P, Q, R, S (both ways)		V / OL
Anode leg; color or type		
1.0 kΩ: V_LED, V_R, I		V, mA
10 kΩ: V_LED, V_R, I		V, mA
Diode S forward: V_S, V_R		V
Diode S reversed: V_S, V_R; why		V

Units and significant figures on every number. Write the meter range next to each reading.

Supervisor sign-off: _____ Score: ____ / 20